

PROJECT AND TEAM INFORMATION

Project Title

Driver Monitoring System : Drowsiness & Mobile Distraction Detection with Mobile Integration

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Driver's drowsiness and use of cellular phone are the main reasons for driving accidents. Long time driving, sleeping less and tiredness decreases the driver's awareness on the road and increases accident rates. Similarly, using or handling a mobile phone while driving can significantly distract the driver and increase the risk of accidents.

The motivation behind this project is to develop an intelligent Driver Monitoring System that can detect signs of drowsiness and mobile distraction in real time. The system uses Computer Vision to monitor the driver's facial features and analyse indicators related to drowsiness, such as eye closure and yawning. To make the detection process more intelligent and data-driven, a Machine Learning approach using a Random Forest Classifier will be incorporated for classifying the driver's condition based on extracted features.

A publicly available drowsiness detection dataset is used to support the development and evaluation of the system. By training and testing the model on relevant data, the project aims to improve the reliability of driver condition classification.

The system also includes mobile phone distraction detection and an alert mechanism with mobile integration. The overall goal is to combine Machine Learning, Computer Vision, and real-time monitoring to create an affordable and practical driver safety solution that can provide timely warnings and potentially help reduce accidents caused by drowsiness and distraction.

State of the Art / Current solution (1 pt)

Currently, driver's drowsiness and distraction are addressed through the use of Advanced Driver Assistance Systems, camera-based monitoring systems, wearable devices, and vehicle behaviour analysis. The camera-based systems usually utilize Computer Vision algorithms to estimate the percentage of eye closure, blink, head pose, and yawning. Others implement Machine Learning and Deep Learning algorithms to classify the driver's status based on facial images or extracted features.

In the case of drowsiness detection, there is a combination of rule-based approaches such as using the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) thresholds for detecting a sleepy driver. More sophisticated Machine Learning algorithms can also be used which involve the use of a database of facial images or extracted features to train a suitable classification model. With respect to mobile distraction, Object Detection algorithms can be used to detect the presence of a mobile phone.

While there are several state-of-the-art driver monitoring systems currently available in the market, they can be very expensive and might require the purchase of additional hardware. Our proposed approach is aimed at providing a prototype of an affordable driver monitoring system based on Computer Vision and a Random Forest Classifier. A publicly available drowsiness detection dataset will be used to train and evaluate the proposed Machine Learning model for classification. In addition to the drowsiness detection, our system will also detect mobile distraction with real-time alerts and mobile integration providing a better overall solution.

Project Goals and Milestones (2 pts)

The main objective of this project is to design an intelligent Driver Monitoring System to track driver's drowsiness and mobile phone distraction in a real-time environment. The proposed system aims to utilize Computer Vision and Machine Learning to monitor the driver and alert them in case of any dangerous activity.

The first major task is to review the existing driver monitoring approaches and finalize the system requirements, technologies, and design/development approach. The second task is to collect and prepare the required dataset for drowsiness detection. In this step, an existing publicly available drowsiness detection dataset will be used for training and testing purposes for the Machine Learning model.

The next task is to extract the relevant features and develop a Random Forest Classifier to classify and predict the driver's state in real-time. The model will be trained and tested to evaluate its performance and accuracy.

The fourth task is to develop the real-time drowsiness detection module using a live camera and Computer Vision techniques. The relevant facial features and other indicators such as closed eyes and yawning will be extracted and processed with the Random Forest Classifier.

The fifth task is to develop the mobile distraction detection module and integrate it with the drowsiness detection module. An alert module will be added to the system to provide audio or visual notification, and mobile notification will also be added.

The last three tasks are to develop all modules into a working application, test and evaluate the performance of the application, improve any weaknesses or limitations, and finalize the project documentation and prepare and demonstrate the project results.

Project Approach (3 pts)

The Driver Monitoring System proposed in this conceptual plan will be created using modules based on Computer Vision, Machine Learning and real-time monitoring. The camera will capture video of the driver and process this data to identify signs of drowsiness and mobile phone distraction.

For the Machine Learning part, a publicly available drowsiness detection dataset will be prepared for processing and testing. Features will be extracted and a Random Forest Classifier will be trained on the data. This model will then be able to classify driver state and indicate if they are showing signs of drowsiness.

To do this in real-time, Python, OpenCV and MediaPipe Face Mesh will be used to extract facial features and calculate Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to detect eye closure and yawning. These can also be used as features for the Machine Learning-based classification.

A separate mobile distraction detection module will be created to detect mobile phone usage while driving. The results from the drowsiness and distraction analysis will then be passed onto a decision making system.

The project will be constructed using Python and libraries such as OpenCV, MediaPipe, Pandas, NumPy, and Scikit-learn. The Random Forest algorithm will be implemented using Scikit-learn and a user interface will be added using Streamlit. An alert and mobile notification system will finally be added for the completed product.

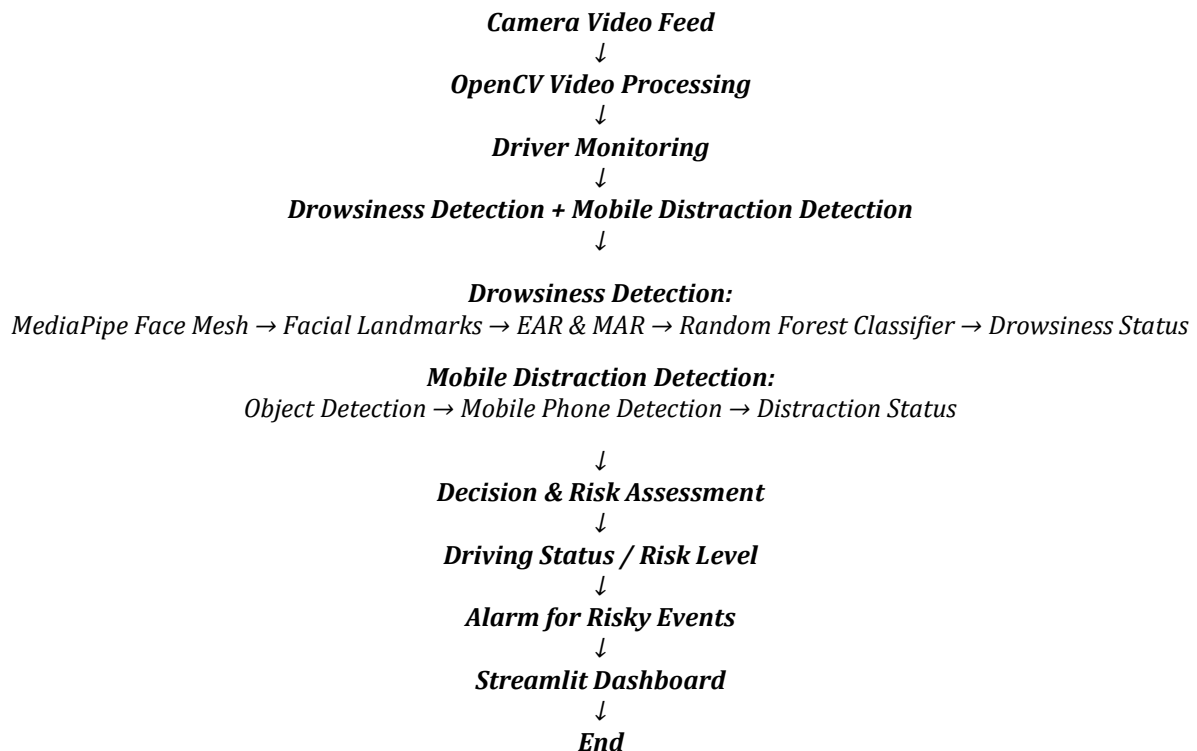
System Architecture (High Level Diagram)(2 pts)

The Driver Monitoring System gets a real-time video feed from a camera and uses OpenCV library for processing and video analysis. The system comprises two modules namely, Drowsiness Detection and Mobile Distraction Detection.

The Drowsiness Detection module employs MediaPipe Face Mesh to detect facial features and extract facial landmarks. It uses geometrical values of facial features such as Eye Aspect Ratio and Mouth Aspect Ratio to identify driving tiredness. The extracted facial features are fed into a Random Forest Classifier. This classification model predicts the status of the driver.

Further, the Mobile Distraction Detection module makes use of an object detection algorithm to detect the usage of a mobile phone. A decision and risk assessment module consolidates the information from the two modules and interprets the driving status of the driver. Based on the interpretation, the system then issues an alarm for risky events such as drowsiness and mobile phone usage. The system also uses mobile integration to alert the user or notify a connected mobile device with the status of the driver. Finally, a dashboard was designed using streamlit to present this application in an interactive web interface.

Driver Monitoring System Flowchart



Project Outcome / Deliverables (1 pts)

The primary output of this project will be a proof-of-concept prototype of the AI-based Driver Monitoring System for the real-time acquisition of the driver's state. This system will leverage a live camera feed to detect the signs of driver drowsiness, such as closed eyes for an extended period of time and a yawn, within the computer vision paradigm. These features will be engineered using the MediaPipe Face Mesh library by computing the relevant geometrical measurements, including Eye Aspect Ratio and Mouth Aspect Ratio. The driver's state will then be classified using A Machine Learning-based approach implemented as a Random Forest Classifier.

In addition, the project will involve developing a mobile distraction detection module that will be able to identify the presence of a mobile phone in the field of view or its active use during the driving session. The system will raise either audio or visual alerts in case of any unsafe driving patterns and will be optionally integrated with a mobile application to receive mobile notifications.

Finally, the project will deliver all the source code developed for the project, including the training pipeline and the Random Forest model, the code for data preparation and feature engineering, and a Streamlit-based dashboard for the user interface and alerts. In terms of documentation, this project will provide the technical specifications of the system, along with the project reports, the results of testing and validation, and the details of the underlying technologies, approaches, and implementations. Overall, the final prototype will provide a viable application of Machine Learning and Computer Vision to monitor the driver and ensure safety.

Assumptions

The proposed system assumes that the camera is correctly positioned to capture the driver's face. Sufficient lighting and visibility of facial features are required for the system to accurately detect landmarks and extract features. The driver must be reasonably visible to the camera. The model based on the Random Forest algorithm assumes that the data used for training will be representative of the data tested. In the case of mobile phone distraction detection, the mobile phone should be visible to the camera. The system is still a prototype, and its performance may be affected by poor lighting, face occlusion, extreme movements, unusual angles, or other factors not present in the data set.

References

- ② [Roboflow Universe – Drowsiness Detection Dataset](#)
- ② [MediaPipe Documentation – Face Landmark Detection](#)
- ② [OpenCV Documentation](#)
- ② [Scikit-learn Documentation – Random Forest Classifier](#)
- ② [Streamlit Documentation](#)
- ② Soukupová, T., & Čech, J. (2016). Real-Time Eye Blink Detection using Facial Landmarks. Czech Technical University in Prague.
- ② [Ultralytics Documentation – YOLO Object Detection](#)